# Experiment 03: Exploratory Data Analysis and Visualization

**Aim:** To perform exploratory data analysis (EDA) and data visualization on the loan dataset using Python.

## Objectives:

1. To load and explore the JioHotstar dataset using Python.
2. To calculate statistical measures of central tendency such as mean, median and mode.
3. To calculate measures of dispersion such as standard deviation, variance, range and interquartile range.
4. To identify relationships between numerical attributes using correlation analysis.
5. To visualize the dataset using histogram, boxplot, correlation heatmap and scatter plot.
6. To identify useful patterns, relationships and variations in the JioHotstar viewing data.

**Theory:** Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset before applying further analysis or machine learning techniques.

**In this experiment, EDA is performed on the preprocessed JioHotstar dataset obtained after Experiment 2.**

The **JioHotstar** dataset contains information related to content characteristics and viewer behaviour, including episode duration, pacing score, hook strength, visual intensity, average watch percentage, pause count, rewind count, cognitive load and drop-off probability. The purpose of EDA is to understand how these attributes are distributed and how they are related to viewer engagement and drop-off.

**EDA is important because it helps in:**

1. **Data Quality Checking:** It helps identify unusual values, variations, and possible outliers in the dataset.
2. **Understanding Data Distribution:** Statistical measures and graphs help understand how numerical attributes are distributed.
3. **Identifying Important Features:** EDA helps identify attributes that may have an important relationship with viewer engagement and drop-off.
4. **Detecting Relationships and Patterns:** Correlation analysis helps determine the strength and direction of relationships between numerical attributes.
5. **Improving Data Understanding:** Visualization makes complex patterns easier to understand and interpret.

**1. Descriptive Analysis – Measures of Central Tendency**

Central tendency measures indicate the **center of the data distribution**.

**It includes:**

**Mean:** Represents the average value of a numerical attribute, such as average episode duration or average watch percentage.

**Median:** Represents the middle value when the observations are arranged in order. It is useful when the data contains extreme values or outliers.

**Mode:** Represents the most frequently occurring value. It is useful for both numerical and categorical attributes, such as platform, genre, retention risk, and watch engagement level.

**These measures help identify the typical characteristics of JioHotstar content and viewer behaviour.**

**2. Descriptive Analysis – Statistical Measures of Dispersion**

Dispersion measures describe the **spread or variability of data**.

It includes:

**Standard Deviation:** Shows how much the observations deviate from the mean.

**Variance:** Quantifies the spread of the data and is the squared value of standard deviation.

**Range:** Represents the difference between the maximum and minimum values.

**Interquartile Range (IQR):** Represents the range of the middle 50% of observations and is useful for identifying potential outliers.

These measures help determine how much attributes such as episode duration, watch percentage, pause count, and cognitive load vary across the dataset.

**3. Correlation Between Attributes**

**Correlation between Attributes** measures the relationship between two characteristics. It splits into two main branches depending on whether your data is numerical or descriptive:

- **Quantitative Attributes (Variables):** Measures numerical relationships using standard correlation coefficients.
- **Qualitative Attributes (Traits):** Measures descriptive traits (e.g., eye color, employment) using the **Theory of Association of Attributes**.

**Key Rules and Outputs**

- **Positive Association:** The presence of one attribute increases the likelihood of the other.
- **Negative Association:** The presence of one attribute decreases the likelihood of the other.
- **Independence:** The attributes have zero relationship and occur together purely by random chance.

- **The Scale:** Results are measured on a scale from **-1 (perfect negative)** to **+1 (perfect positive)**, where **0** means no relationship.

**4. Different Visualization Techniques & Their Uses**

**Breakdown of Techniques & Strategic Uses;**

**Histogram**

- Primary Use: Distribution and spread analysis.
- How it works: Groups continuous numerical data into ranges (bins) to show frequency density.
- Your Example: *ApplicantIncome* reveals if wealth is concentrated at lower levels, normal, or skewed.

**Boxplot (Box-and-Whisker)**

- Primary Use: Comparative distribution and anomaly detection.
- How it works: Displays the 5-number summary (minimum, first quartile, median, third quartile, maximum) and flags isolated data points.
- Your Example: *LoanAmount by Loan_Status* visually tests if rejected loans have a significantly higher median or more extreme outliers.

**Heatmap**

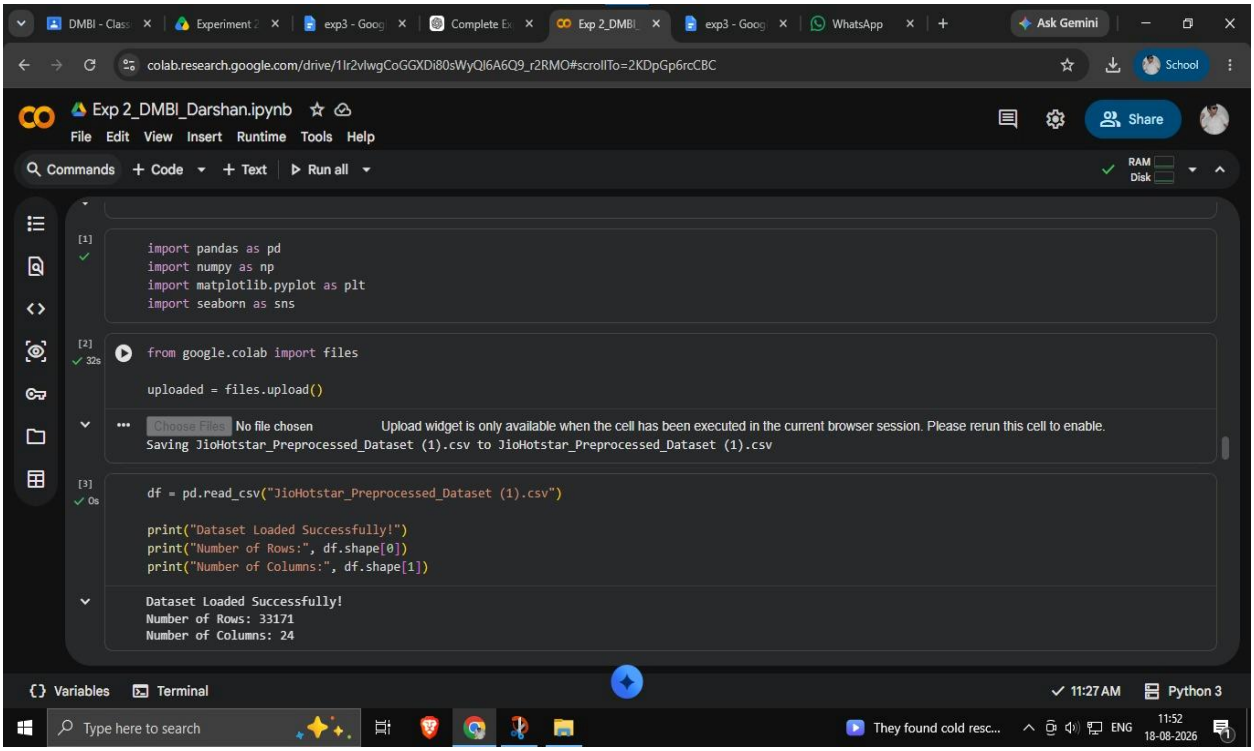
- Primary Use: Multi-variable correlation matrices.
- How it works: Uses a color-coded matrix where shades reflect the strength of linear relationships between feature pairs.
- Your Example: Instantly flags high-dependency clusters (e.g., if *Credit_History* strongly correlates with *Loan_Status*).

**Scatter Plot**

- Primary Use: Bivariate relationship analysis.
- How it works: Plots individual data points across two continuous axes to reveal clusters, trends, or non-linear patterns.
- Your Example: *ApplicantIncome vs. LoanAmount* checks if higher-earning applicants naturally request larger loans, or if requests cap out.

# Codes and Outputs:

## Step 1: Loading the Dataset and Initial Exploration




```
print("First 5 Rows:")
display(df.head())
```

First 5 Rows:

| | show_id | title | platform | genre | release_year | season_number | episode_number | episode_duration_min | pacing_score | hook_strength | ... | rewind_count |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 66732 | Stranger Things | Jio_hotstar | Sci-Fi & Fantasy | 2016.0 | 1 | 1 | 48 | 4 | 5 | ... | 0 |
| 1 | 66732 | Stranger Things | Jio_hotstar | Sci-Fi & Fantasy | 2016.0 | 1 | 2 | 55 | 5 | 4 | ... | 3 |
| 2 | 66732 | Stranger Things | Jio_hotstar | Sci-Fi & Fantasy | 2016.0 | 1 | 3 | 51 | 4 | 8 | ... | 2 |
| 3 | 66732 | Stranger Things | Jio_hotstar | Sci-Fi & Fantasy | 2016.0 | 1 | 4 | 50 | 4 | 7 | ... | 1 |
| 4 | 66732 | Stranger Things | Jio_hotstar | Sci-Fi & Fantasy | 2016.0 | 1 | 5 | 52 | 4 | 3 | ... | 0 |

5 rows × 24 columns

```
print("Columns in Dataset:")
print(df.columns.tolist())
```

```
Columns in Dataset:
['show_id', 'title', 'platform', 'genre', 'release_year', 'season_number', 'episode_number', 'episode_duration_min', 'pacing_score', 'hook_strength', 'd
```

**Step 2: Task 1 - Descriptive Analysis - Statistical Measures of Central Tendency**

```
[7]
✓ 0s
```

```python
# Task 1: Descriptive Analysis - Measures of Central Tendency

numerical_columns = [
    'episode_duration_min',
    'pacing_score',
    'hook_strength',
    'visual_intensity',
    'avg_watch_percentage',
    'pause_count',
    'rewind_count',
    'cognitive_load',
    'drop_off_probability'
]

print("MEAN:")
print(df[numerical_columns].mean())

print("\nMEDIAN:")
print(df[numerical_columns].median())

print("\nMODE:")
print(df[numerical_columns].mode().iloc[0])
```

```
MEAN:
episode_duration_min    37.660667
pacing_score             5.397908
hook_strength            5.525158
visual_intensity         6.009436
pause_count              3.102921
rewind_count             2.003497
cognitive_load           6.196015
drop_off_probability     0.478433
dtype: float64

MEDIAN:
episode_duration_min    33.000
pacing_score             5.000
hook_strength            6.000
visual_intensity         6.000
avg_watch_percentage    57.000
pause_count              3.000
rewind_count             2.000
cognitive_load           7.000
drop_off_probability     0.471
dtype: float64

MODE:
episode_duration_min    25.00
pacing_score             6.00
hook_strength            6.00
visual_intensity         6.00
avg_watch_percentage    60.00
pause_count              2.00
rewind_count             1.00
cognitive_load           5.00
drop_off_probability     0.43
Name: 0, dtype: float64
```

**Inferences:**

The mean episode duration is 37.66 minutes, while the median is 33 minutes and the mode is 25 minutes. The difference between the mean and median indicates variation in episode duration, with some longer episodes increasing the average.

The pacing score has a mean of 5.40, median of 5, and mode of 6, indicating generally moderate pacing across the dataset. The hook strength has a mean of 5.53, median of 6, and mode of 6, indicating that relatively strong hooks are common in the dataset.

**Visual intensity has** a mean of approximately 6.01, median of 6, and mode of 6, indicating relatively consistent visual intensity.

**Average watch percentage has** a mean of 56.81%, median of 57%, and mode of 60%, indicating that viewers typically watch around half or slightly more than half of an episode.

Pause count has a mean of 3.10, median of 3, and mode of 2, while rewind count has a mean of approximately 2.00, median of 2, and mode of 1, indicating moderate viewer interaction with the content.

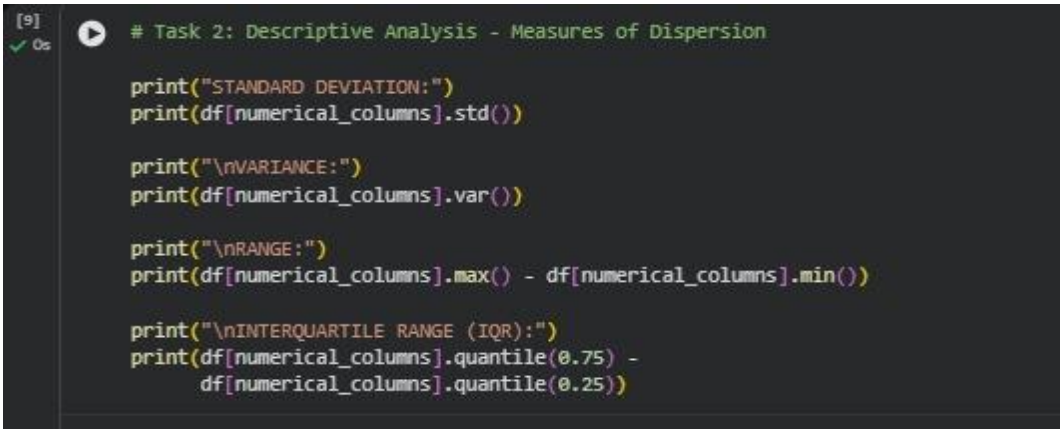
**Cognitive load has** a mean of 6.20, median of 7, and mode of 5, showing variation in the cognitive demand of episodes.

Drop-off probability has a mean of 0.478, median of 0.471, and mode of 0.43, indicating a moderate level of predicted viewer drop-off.

**Insight:**

**The central tendency analysis indicates that the typical JioHotstar episode has moderate pacing, relatively strong hooks, visual intensity around 6, and an average watch percentage of approximately 57%. Viewer interactions through pauses and rewinds are also present, while drop-off probability remains moderate.**

**Step 3: Task 2 - Descriptive Analysis - Statistical Measures of Dispersion**



```
STANDARD DEVIATION:
episode_duration_min    22.899741
pacing_score             1.230418
hook_strength            1.730098
visual_intensity         1.994370
avg_watch_percentage    12.577000
pause_count              1.866938
rewind_count             1.548391
cognitive_load           1.322146
drop_off_probability     0.107382
dtype: float64

VARIANCE:
episode_duration_min    524.398145
pacing_score              1.513929
hook_strength             2.993240
visual_intensity          3.977511
avg_watch_percentage    158.180933
pause_count               3.485458
rewind_count              2.397516
cognitive_load            1.748071
drop_off_probability      0.011531
dtype: float64
```

```
RANGE:
episode_duration_min    1224.000
pacing_score               5.000
hook_strength              7.000
visual_intensity           6.000
avg_watch_percentage      87.000
pause_count               10.000
rewind_count               8.000
cognitive_load             5.000
drop_off_probability       0.711
dtype: float64

INTERQUARTILE RANGE (IQR):
episode_duration_min    25.000
pacing_score             2.000
hook_strength            3.000
visual_intensity         4.000
avg_watch_percentage    18.000
pause_count              2.000
rewind_count             2.000
cognitive_load           2.000
drop_off_probability     0.153
dtype: float64
```

**Insight:** The dispersion analysis shows that episode duration and average watch percentage have relatively high variability, indicating significant differences in episode length and viewer watch behaviour. Pacing score and cognitive load show comparatively lower variation, suggesting that these attributes are more consistent across the dataset. The IQR values provide the spread of the middle 50% of observations and can be useful for identifying potential outliers.

## Step 4: Task 3 - Correlation Between Attributes

```
[10] ✓ 0s
# Task 3: Correlation Between Numerical Attributes

correlation_matrix = df[numerical_columns].corr()

print("Correlation Matrix:")
display(correlation_matrix.round(2))
```

Correlation Matrix:

| | episode_duration_min | pacing_score | hook_strength | visual_intensity | avg_watch_percentage | pause_count | rewind_count | cognitive_load | drop_off_probability |
|---|---|---|---|---|---|---|---|---|---|
| episode_duration_min | 1.00 | -0.01 | 0.01 | -0.01 | -0.00 | 0.00 | 0.01 | 0.00 | -0.00 |
| pacing_score | -0.01 | 1.00 | -0.00 | -0.00 | 0.59 | -0.19 | -0.00 | -0.57 | -0.56 |
| hook_strength | 0.01 | -0.00 | 1.00 | -0.01 | 0.55 | 0.00 | -0.01 | 0.00 | -0.53 |
| visual_intensity | -0.01 | -0.00 | -0.01 | 1.00 | -0.01 | 0.01 | 0.42 | 0.01 | 0.01 |
| avg_watch_percentage | -0.00 | 0.59 | 0.55 | -0.01 | 1.00 | -0.24 | -0.01 | -0.69 | -0.96 |
| pause_count | 0.00 | -0.19 | 0.00 | 0.01 | -0.24 | 1.00 | 0.01 | 0.34 | 0.42 |
| rewind_count | 0.01 | -0.00 | -0.01 | 0.42 | -0.01 | 0.01 | 1.00 | 0.00 | 0.01 |
| cognitive_load | 0.00 | -0.57 | 0.00 | 0.01 | -0.69 | 0.34 | 0.00 | 1.00 | 0.79 |
| drop_off_probability | -0.00 | -0.56 | -0.53 | 0.01 | -0.96 | 0.42 | 0.01 | 0.79 | 1.00 |

**Insight:**

**The correlation analysis shows that average watch percentage has a very strong negative relationship with drop-off probability (-0.96), indicating that higher watch percentage is associated with lower drop-off probability. Cognitive load has a strong positive relationship with drop-off probability (+0.79) and a negative relationship with average watch percentage (-0.69). Pacing score and hook strength also show moderate relationships with viewer engagement. Overall, average watch percentage, cognitive load, pacing score, and hook strength appear to be important attributes for understanding viewer retention and drop-off.**

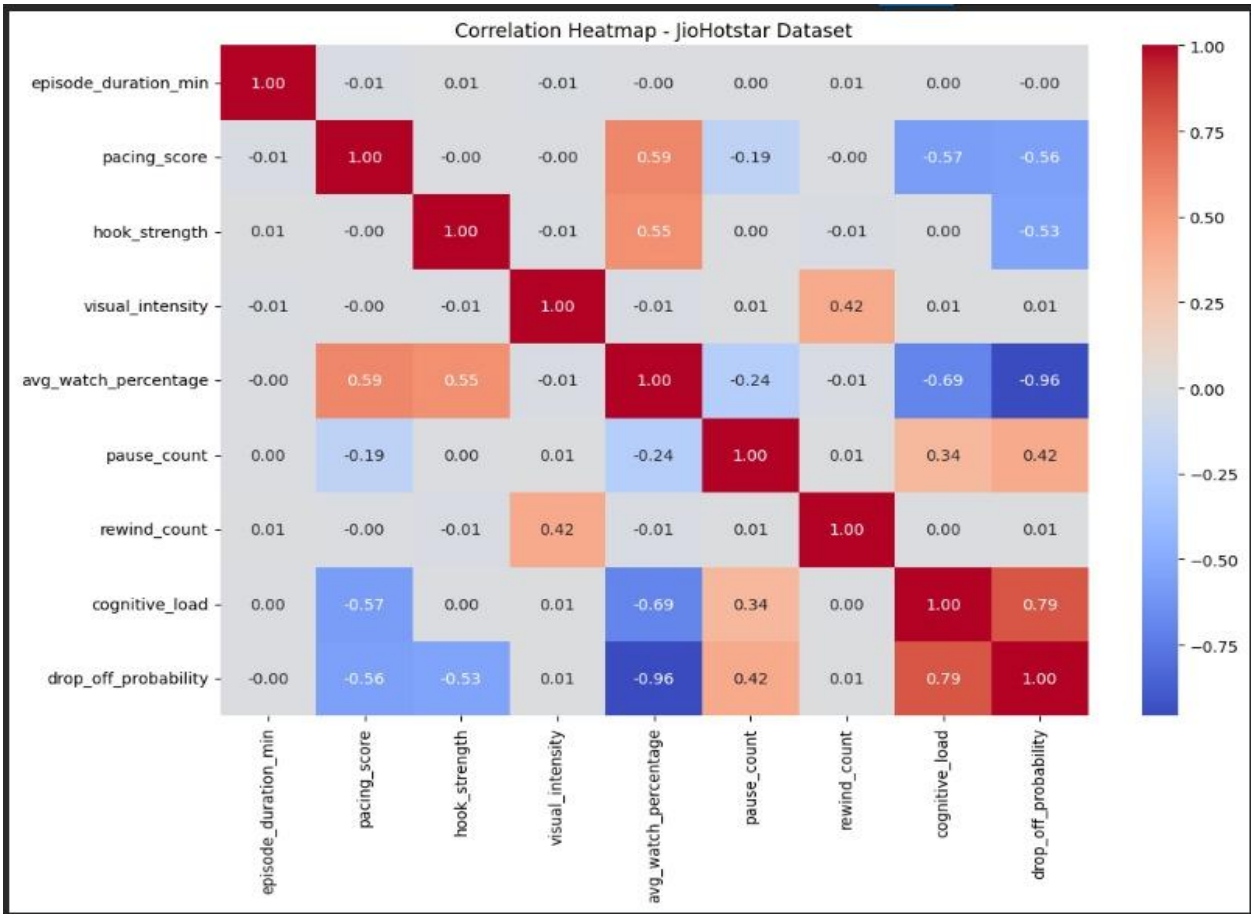
**Different Visualization Techniques:**

```
[12] ✓ 0s
# Task 3B: Correlation Heatmap

plt.figure(figsize=(12, 8))

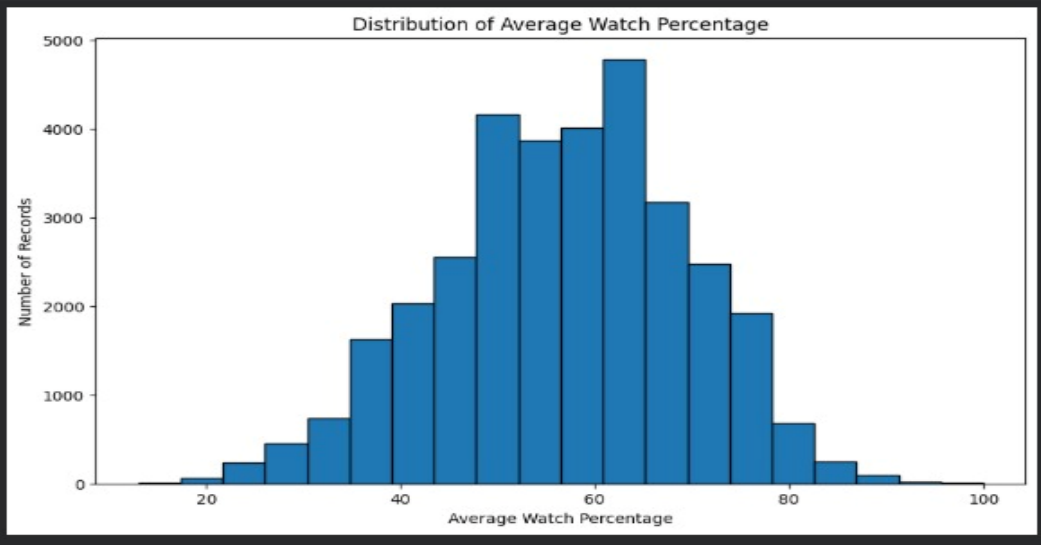
sns.heatmap(
    correlation_matrix,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap - JioHotstar Dataset")
plt.show()
```




```
plt.hist(
    df['avg_watch_percentage'],
    bins=20,
    edgecolor='black'
)

plt.xlabel("Average Watch Percentage")
plt.ylabel("Number of Records")
plt.title("Distribution of Average Watch Percentage")
plt.show()
```

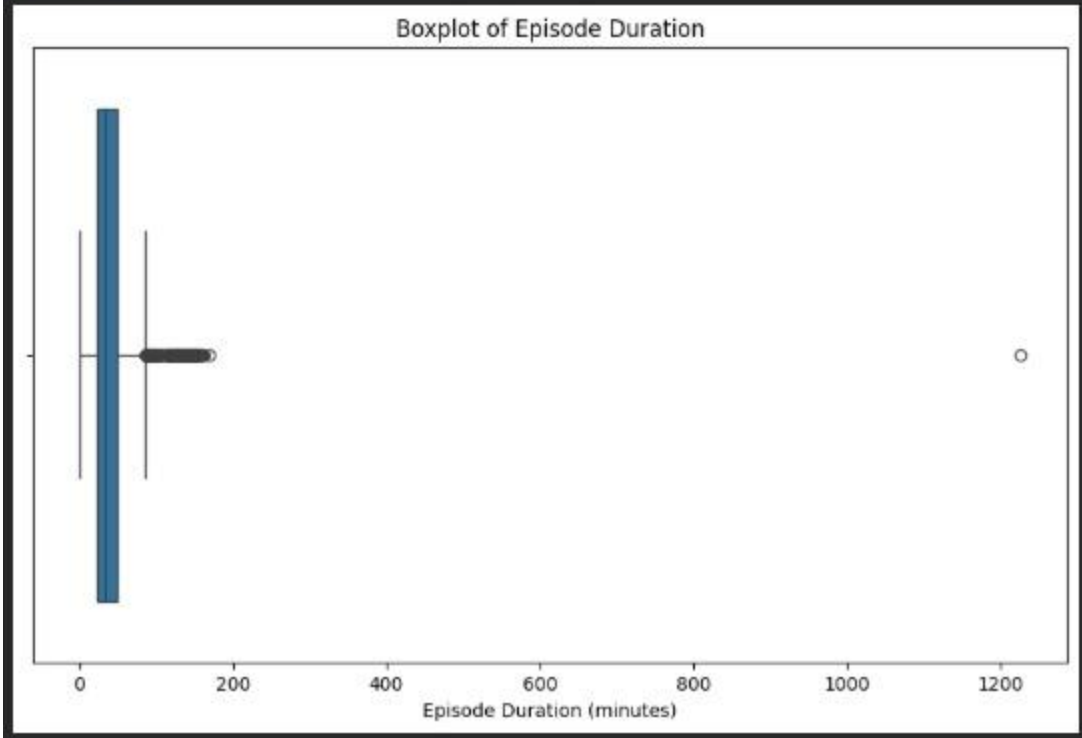


```python
# Task 4B: Boxplot of Episode Duration

plt.figure(figsize=(10, 6))

sns.boxplot(x=df['episode_duration_min'])

plt.xlabel("Episode Duration (minutes)")
plt.title("Boxplot of Episode Duration")
plt.show()
```



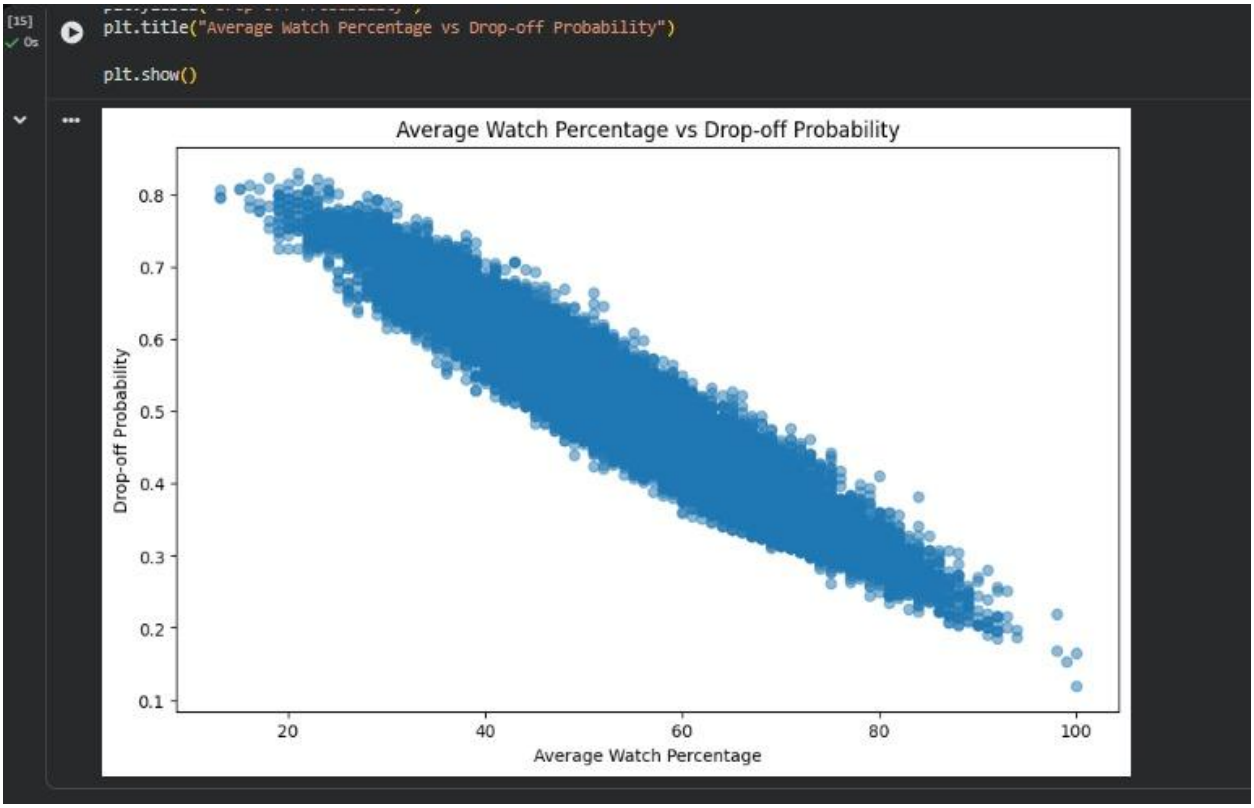

```python
# Task 4C: Scatter Plot

plt.figure(figsize=(10, 6))

plt.scatter(
    df['avg_watch_percentage'],
    df['drop_off_probability'],
    alpha=0.5
)

plt.xlabel("Average Watch Percentage")
plt.ylabel("Drop-off Probability")
plt.title("Average Watch Percentage vs Drop-off Probability")

plt.show()
```




**1. Histogram — Average Watch Percentage**

**Inference:**

The histogram represents the distribution of average watch percentage across the JioHotstar dataset. It helps identify the range in which most observations are concentrated and shows the overall variation in viewer watch behaviour. The mean and median values of approximately 56.81% and 57% indicate that the typical watch percentage is around the mid-50% range.

**Insight:**

The histogram provides a clear view of viewer watch-percentage distribution and helps understand the typical level of content watched by viewers.

**2. Boxplot — Episode Duration**

**Inference:**

The boxplot represents the distribution of episode duration using the median, quartiles, and whiskers. It helps identify the spread of episode durations and potential outliers. The large range of 1224 minutes and standard deviation of approximately 22.90 minutes indicate considerable variation in episode duration.

**Insight:** The boxplot shows that episode duration varies considerably across the dataset, and any observations beyond the whiskers can be considered potential outliers requiring further examination.

**3. Correlation Heatmap**

**Inference:**

The correlation heatmap provides a visual representation of the relationships between numerical attributes in the JioHotstar dataset. A very strong negative correlation is observed between average watch percentage and drop-off probability (-0.96), while a strong positive correlation is observed between cognitive load and drop-off probability (+0.79). Cognitive load and average watch percentage also show a negative correlation of -0.69.

**Insight:**

The heatmap makes strong positive and negative relationships easier to identify visually. It highlights average watch percentage and cognitive load as important attributes associated with viewer drop-off.

**4. Scatter Plot — Average Watch Percentage vs Drop-off Probability**

**Inference:**

The scatter plot shows a clear negative relationship between average watch percentage and drop-off probability. As the average watch percentage increases, the drop-off probability generally decreases. This agrees with the correlation coefficient of -0.96, indicating a very strong negative relationship.

**Insight:**

The scatter plot confirms that higher viewer watch percentage is strongly associated with lower drop-off probability, making average watch percentage an important indicator of viewer retention.

## Conclusion:

The Exploratory Data Analysis (EDA) and Visualization of the JioHotstar dataset was successfully performed using Python. The analysis included measures of central tendency, measures of dispersion, correlation analysis, and different visualization techniques such as histogram, boxplot, correlation heatmap, and scatter plot.

The analysis showed that episode duration and average watch percentage have considerable variation across the dataset. The correlation analysis revealed a **very strong negative correlation (-0.96) between average watch percentage and drop-off probability**, indicating that higher watch percentage is strongly associated with lower drop-off probability. **A strong positive correlation (+0.79) was also observed between cognitive load and drop-off probability.**

**The visualizations helped in understanding the distribution, variability, relationships, and potential outliers present in the JioHotstar data.** Overall, the experiment demonstrates that EDA is useful for identifying important patterns in viewer engagement and retention behaviour and provides a foundation for further data analysis and predictive modelling.